

1	(a)	$\theta \text{ (rad)} = 2\pi \times (10.3/360)$	1	[2]
		$= 0.180 \text{ rad}$ (n.b. 3 sig. fig.)	1	

	(b)	(i)	$\tan \theta = 0.182$	1	
			(n.b. 3 sig. fig.)		

		(ii)	percentage error = $(0.002/0.180) \times 100$	1	
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			$= 1.1 \text{ (\%)}$	1	[3]
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(allow 0.002/0.182 and allow 1 → 4 sig. fig.)

2	(a)	(i)	grav. pot. energy = GMm/R	1	
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